Call for Abstracts

SETAC North America 47th Annual Meeting

November 2-5, 2026 - Montréal, Québec, Canada

Session title

Advancing Non-Targeted Analysis: Integrating Analytical Rigour, Informatics Innovation, and Risk Assessment for Exposure Characterization

Session description

Non-targeted analysis (NTA) based on high-resolution mass spectrometry has emerged as a transformative approach in environmental and exposure science, enabling the detection and characterization of thousands of chemicals across complex mixtures in environmental and biological matrices. As NTA methods mature and expand across disciplines, two interconnected priorities must be addressed: the development of robust, harmonized analytical workflows and the informatics infrastructure needed to convert rich experimental datasets into defensible, reproducible chemical identifications.

The first focus of this session addresses recent advances in NTA designed to bridge analytical chemistry, exposure assessment, and risk assessment. Particular emphasis is placed on methodological rigour and harmonization of NTA workflows, including quality assurance and quality control practices, optimization of acquisition strategies, and semi-quantitative and comparative approaches that support chemical prioritization within complex mixtures. Applications across diverse exposure contexts — including urban environments, large river systems, and aquatic and terrestrial ecosystems worldwide — illustrate how NTA captures temporal and spatial variability in real-world exposures. Emerging approaches coupling NTA data with biological, ecotoxicological, and multi-omics information to link chemical exposure to biological responses are also addressed.

The second focus highlights the informatics ecosystem essential to modern NTA. Transforming raw chemical signals into confirmed, probable, or tentative structural identifications requires spectral libraries, structure prediction models, physicochemical property estimates, and stable cheminformatics pipelines. The development of informatics tools — ranging from feature annotation frameworks and machine learning models to cloud-based platforms and database-driven pipelines — has the potential to dramatically improve the speed, confidence, and reproducibility of NTA results. Presentations highlighting novel platforms for handling high-dimensional NTA datasets, tools integrating experimental and in silico data, and automated workflows enhancing transparency and reproducibility are featured.

By bringing together developers, practitioners, analytical chemists, exposure scientists, and risk assessors, this session demonstrates how NTA is evolving from an exploratory tool into a robust, transparent, and scalable component of environmental decision-making.

We welcome contributions on

- Harmonized NTA workflows and quality assurance/quality control practices
- Semi-quantitative and comparative approaches for chemical prioritization
- Applications of NTA in environmental and exposure contexts
- Coupling NTA with biological, ecotoxicological, and multi-omics data
- Informatics tools, spectral libraries, and cheminformatics pipelines
- Machine learning and AI-driven approaches for compound identification
- Cloud-based and automated workflows for NTA data processing

- Integration of NTA into risk assessment frameworks

Presentation types

- Both oral presentations and posters are welcome.

Key dates

- **Abstract submission deadline (free):** June 3, 2026
- **Abstract submission deadline ($50 fee):** June 10, 2026
- **Meeting registration deadline:** October 14, 2026
- **Annual Meeting:** November 2–5, 2026 — Montréal, Québec

Session chairs

- **Morgan Solliec,** MELCCFP
- **Stéphane Bayen,** McGill University
- **Nathaniel Charest**, ULRI
- **Heather Whitehead**, ULRI

How to submit

- Submit your abstract at: **setac.org/montreal**
- For questions, contact: **morgan.solliec@environnement.gouv.qc.ca** ; **north-america-meeting@setac.org**